# HEAR MY HANDS

## A Python-Based Real-Time Sign Language Interpreter Using Computer Vision and Machine Learning

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## Introduction

Hear My Hands is designed to use a webcam or mobile camera to recognize at least 50 common signs and convert them into text and speech in real time. The project aims for 85% accuracy and a response time of less than four seconds. The system is based on Python, OpenCV, Media Pipe, and CNNs, and it follows a workflow from camera input to sign recognition output.

Researchers do not treat sign language recognition as a simple gesture classification problem. This is because sign languages use both manual features, such as hand shape and movement, and non-manual features, such as facial expressions and body posture. In addition, successful sign language systems need to consider language, Deaf culture, accessibility, and human-computer interaction, not just computer vision techniques (Bragg et al., 2019; Liang et al., 2023; Rastgoo et al., 2021). Research on Turkish sign language also shows that signs involve hand orientation, movement, posture, and facial information at the same time. This means that systems that only look at cropped hand images may miss important information (Sincan & Keles, 2020).

# Theoretical Background

## Artificial Intelligence and Machine Learning

While machine learning (ML) enables systems to recognize patterns in data, artificial intelligence (AI) enables systems to mimic human intelligence. In order to correctly classify and interpret signs, machine learning models are trained using datasets of hand gestures.

## Computer Vision

In order to capture and process visual input from cameras, computer vision is essential. Real-time hand shape, movement, and position identification is achieved through methods like image processing, object detection, and feature extraction.

## Deep Learning Techniques

Gesture recognition makes extensive use of deep learning models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). While RNNs are helpful for sequential gesture interpretation, CNNs aid in the recognition of spatial features, such as hand shapes.

## Critical Analysis and Synthesis

An important idea in the literature is the difference between isolated sign recognition, continuous sign recognition, and sign language translation. Isolated sign recognition identifies one sign from one video clip. Continuous sign recognition identifies a sequence of signs over time. Sign language translation goes one step further by converting signs into spoken-language sentences, where the grammar and word order may be different from the original sign language (Liang et al., 2023; Camgoz et al., 2018; Camgoz et al., 2020). Since

Hear My Hands focuses on 50 common signs and is measured mainly by accuracy and speed, it is more suitable as an isolated sign recognition system rather than a full translation system, even though it produces text and speech output (Liang et al., 2023; Camgoz et al., 2020). Previous studies on large continuous sign language datasets, such as RWTH-PHOENIX-Weather, show that continuous recognition requires much larger labelled datasets and more complex temporal models than a short project usually allows (Forster et al., 2014; Cui et al., 2019).

## Advancements in Sign Language Recognition

Another major trend in the literature is the shift from special sensors to regular RGB camera systems. For example, BosphorusSign was created using Kinect v2 depth sensing, and AUTSL includes RGB, depth, and skeleton data. On the other hand, MS-ASL and WLASL show that large real-world RGB datasets can support signer-independent recognition using more practical and accessible hardware (Camgoz et al., 2016; Sincan & Keles, 2020; Vaezi Joze & Koller, 2018; Li et al., 2020). This supports the decision in Hear My Hands to use a webcam or mobile camera. However, it also means the system must deal with more variation in lighting, background, camera angle, and signing style compared with systems built in controlled environments (Vaezi Joze & Koller, 2018; Sincan & Keles, 2020; Li et al., 2020).

The development of sign language models has also changed over time. Earlier systems often used handcrafted features, but recent work has moved toward deep learning

methods that can learn both spatial and temporal patterns. These include recurrent models for continuous recognition and transformer-based models for recognition and translation, because many signs depend on movement over time and not just on a single hand shape (Cui et al., 2019; Camgoz et al., 2020; Rastgoo et al., 2021). For a low-cost real-time project, landmark-based methods are still useful. MediaPipe Hands can track hand landmarks in real time from a single RGB camera, which reduces the amount of work needed for the recognition model (Zhang et al., 2020). For this reason, the use of Python, OpenCV, MediaPipe, webcam input, feature extraction, and a CNN-based recognition stage in Hear My Hands is in line with current practice. At the same time, the literature suggests that adding temporal models and including body or face information could improve the system beyond simple frame-by-frame classification (Zhang et al., 2020; Sincan & Keles, 2020; Rastgoo et al., 2021).

Another common finding in the literature is that the quality and diversity of data often affect performance more than the choice of model itself, especially when systems are tested on new users and in realistic situations (Vaezi Joze & Koller, 2018; Li et al., 2020; Sincan & Keles, 2020). Our project also identifies low accuracy, limited training data, and camera limitations as key risks. This suggests that collecting data from multiple signers, keeping the classes balanced, and testing the system in realistic conditions should be major priorities. The literature therefore supports building a local dataset for the selected 50 signs with variation in clothing, lighting, camera angle, background, and signer identity if the project wants to achieve 85% accuracy outside a controlled lab setting (Vaezi Joze & Koller, 2018; Li et al., 2020; Sincan & Keles, 2020).

The literature also makes it clear that sign languages are not universal. Different public datasets are usually built for specific sign languages, such as American Sign Language, German Sign Language, and Turkish Sign Language, rather than for a single global sign language system (Bragg et al., 2019; Forster et al., 2014; Sincan & Keles, 2020; Li et al., 2020). We will be using the Australian Sign Language vocabulary. This is important because the vocabulary, annotations, and benchmark comparisons all depend on the chosen sign language. Domain-specific datasets such as BosphorusSign also show that a limited vocabulary in areas such as health, finance, or daily communication can be a practical starting point for assistive technology, which fits well with the project's proposed use in community, education, and healthcare settings (Camgoz et al., 2016).

## Conclusion

Overall, the literature suggests that Hear My Hands is a realistic first-stage assistive technology project if it is treated as a vision-based isolated sign recognition system that uses standard RGB cameras, real-time hand landmark extraction, and a lightweight temporal classifier to produce text and speech for a clearly defined vocabulary and target sign language (Liang et al., 2023; Zhang et al., 2020; Li et al., 2020). This design fits the project's low-cost hardware plan, 13-week development timeline, and clear performance targets. It also provides a strong foundation for future work on larger vocabularies, continuous recognition, and full sign language translation when more time and more diverse data become available (Camgoz et al., 2020; Forster et al., 2014).

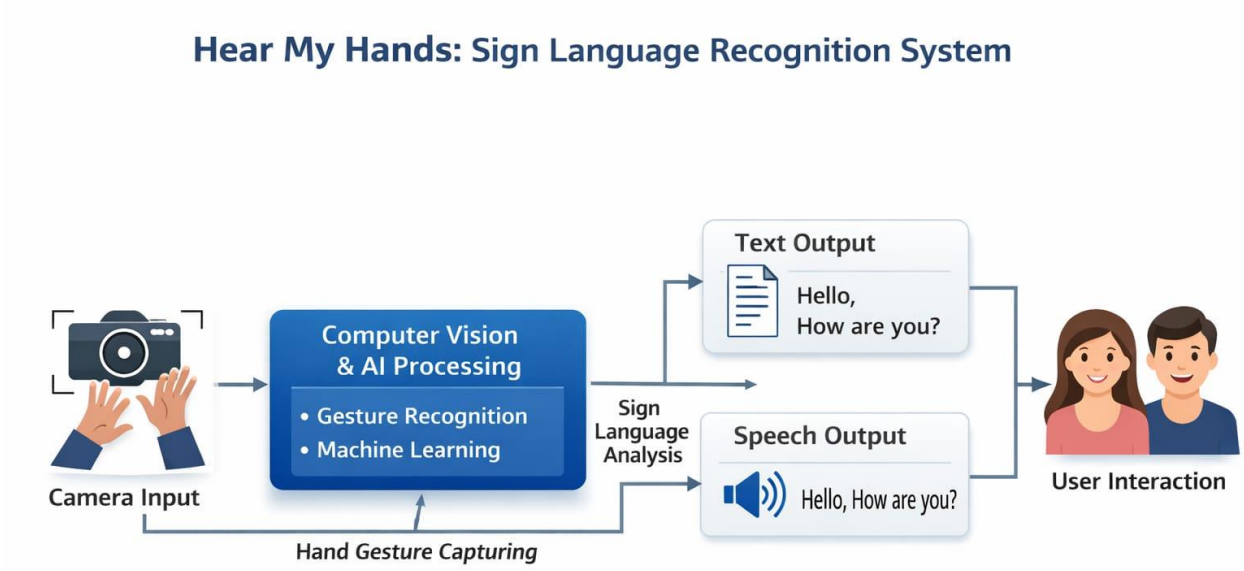



*Figure 1: Block Diagram of Hear My Hands*

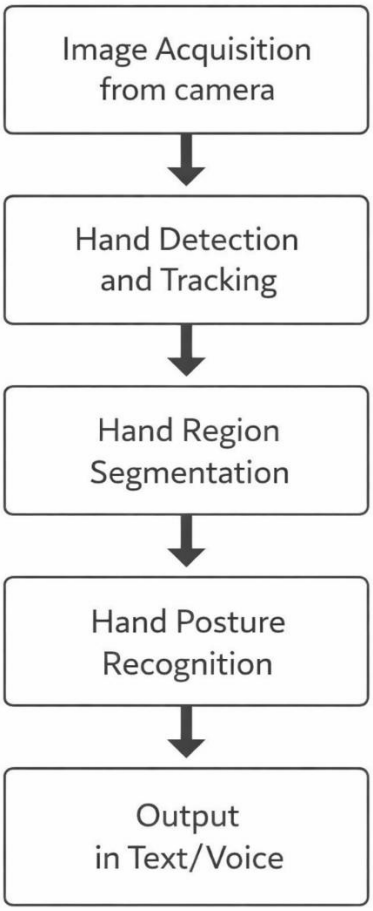



*Figure 2: Dataset Process*

## Table 1

### Success Rate of ASL Alphabets

| Alphabet | Success Rate (%) |
|---|---|
| A | 100 |
| B | 100 |
| C | 100 |
| D | 100 |
| E | 100 |
| F | 85.00 |
| G | 100 |
| H | 100 |
| I | 100 |
| IJ | 100 |
| L | 85.00 |
| M | 85.00 |
| N | 100 |
| O | 100 |
| P | 84.00 |
| Q | 79.00 |
| R | 100 |
| S | 100 |
| U | 89.00 |
| V | 87.00 |
| X | 89.00 |
| Y | 100 |
| Z | 99.00 |

*Figure 3:Succes Rate based on other research.*

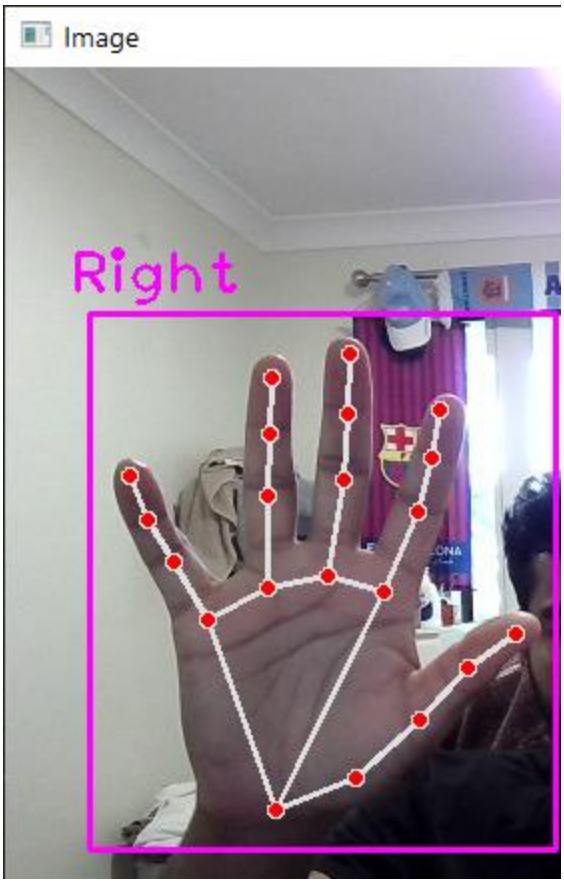



*Figure 4: System Detecting Right Hand Movement*

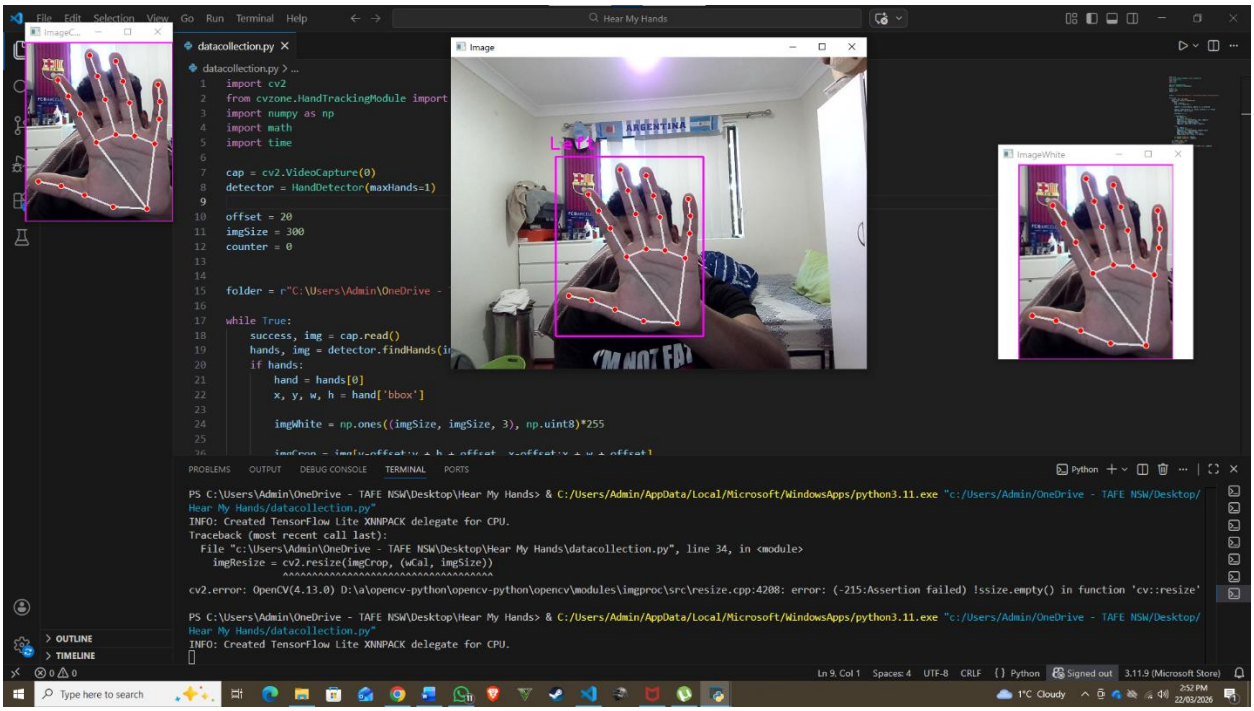



*Figure 5: Sign Language Intrepretor Interface*

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